$L(x) = \int \text{gradient}\, dx = \int (x^3 - x)\, dx = \frac{1}{4}x^4 - \frac{1}{2}x^2 \cancel{+C} = L(x)$ "loss"

$= \frac{1}{4}\left(\frac{1}{2}x^2 - 1\right)^2$

$+, = x_n - \nabla L(x_n) = 2x_n - x_n^3$

$\{ x_n^3 - x_n = \nabla L(x_n)$ "gradient"

[0x24]

$x_0 \cdot x_k \le 1$

ie $\min f(x) = \frac{1}{4}\|x^2 - 1\|_2^2$ OR

ie "obj. fn" w/ tolerance check of: $\text{rax}_0 \cdot \text{rax}_n \le 1$

$-\text{rax}^3 + 2\text{rax}$

$f(x) = \text{rax} \cdot -(\text{rax}^2 - 2)$ $x_k = x_{k-1} \cdot -(x_{k-1}^2 - 2)$

ie $x_{k+1} = x_k \cdot -(x_k^2 - 2)$ (OR

x^2 → next

initial guess?? divide/guess: idempotent/mod?

```
                mov     rbx, rax        rbx, rsi := rax
                mov     rsi, rax
rbx:    .loop:
rcx = rbx·rax   mul     rbx             upper  lower  128-bit value
                mov     rcx, rax        rdx:rax := rbx·rax
                                        rcx

rax:
rsi = rsi·-(rax-2)  sub  rax, 2         rax = rax - 2 + CF
                    neg  rax            rax = -rax
                    mul  rsi            rdx:rax = rsi·rax
rsi = rax·-(rax² - 2)  mov  rsi, rax    rsi := rax

                cmp     rcx, 1
                ja      .loop           ⇒ while(rcx > 1)
                .exit_loop:
```

polynomial optimization over?

(b), (5) = (a)

rbx, rsi := rax

(c)

while (rcx > 1)

{

$(a)_n = (b)(a)_{n-1} = c$

$(\text{rdx:rax})_n := \text{rbx} \cdot \text{rax} \approx \text{rcx}$] tolerance OR convergence check

rcx := rax

$((a)_{n-1} := 2 + CF)$

rax := rax - 2 + CF

rax = -rax

rdx:rax := rsi·rax

rsi := rax

}

loop

rax:

$\text{rax}_{n-1} \cdot \text{rax}_n$

rbx

current

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